Autonomous Mowers

A cost and feasibility review for Greeneville City Schools grounds maintenance

Prepared for Jason B. Horne, Assistant Director of Schools for Administration | August 22, 2026 | Research review, not a purchase recommendation

1. Executive Summary

Autonomous mowing is no longer experimental. As of 2026 there are three distinct product classes on the market, all with real customers, real dealers, and published prices. The question for GCS is not whether the technology works. It is whether GCS has enough contiguous, low-obstacle turf to make the math work, and whether the district wants to convert grounds labor into a fixed equipment cost.

The short version of the finding is this. Autonomous mowing pays back fastest on large, flat, open, fenced turf that currently consumes a lot of repetitive staff hours. It pays back slowest on small, hilly, tree-covered, obstacle-dense lots that need a person with a trimmer anyway. GCS almost certainly has both kinds of ground. The savings case lives on the athletic complexes and the large open lawns, not on the whole district at once.

What the money looks like

Item	Range found	Source type
Light commercial robot (3 to 4.5 acres)	$8,500 per unit	Dealer list price
Mid commercial robot (6 to 9 acres)	$13,800 per unit	Dealer list price
Large commercial robot (up to 18 acres)	$32,830 starter kit	Trade association survey
Autonomous ride-on class	$34,000 to $60,000+	Trade association and dealer
Site setup and mapping	$500 to $3,000+	Dealer guide
Robots as a service (per acre mowed)	about $30 per acre	Trade press
Conventional contract mowing, 15+ acre sites	$35 to $100 per acre per cut	Industry pricing guide
Reported annual savings per site	$5,000 to $15,000+	Vendor-adjacent guide
Reported payback	1.5 to 4 years	Vendor-adjacent guides

Full citations are in Section 11. Source type is stated because reliability varies sharply in this market. See Section 2 for how to weight each kind.

Bottom line up front

- The technology is mature enough to buy, but the savings claims in circulation are mostly vendor-produced and should not be used in a board presentation without district-specific validation.
- The only rigorous, third-party-documented school figure found is Del Mar High School, where one robot on a 1.2 acre front lawn cut mowing labor from 26 to 30 staff hours per week at peak season down to about 6 hours per week of oversight.
- GCS cannot compute a real payback yet. Mowable acreage by site is not recorded anywhere in district files. That single number drives the entire model.

- The most defensible first move is a one-season, one-site pilot on the largest open turf area GCS owns, funded under the open-market purchasing threshold, with hours logged before and after.

2. How to Read the Evidence in This Market

This category has a serious source-quality problem. Most published savings claims trace back to manufacturers, dealers, or affiliate review sites that earn commission. Two of the most widely repeated figures, that robotic mowing drops cost from $40 per acre to $10 per acre and that robotic systems cut long-term costs by 50 to 70 percent, appear in vendor marketing and vendor-adjacent blogs with no underlying methodology published. They are not usable as planning figures.

For this review, evidence is graded in three tiers.

Tier	What it is	How to use it
Tier 1	Public board records, government reporting, independent journalism with named districts and dollar figures	Usable in a board presentation as stated
Tier 2	Published dealer list prices and manufacturer specifications	Usable for cost inputs, verify with a local quote
Tier 3	Vendor blogs, affiliate review sites, manufacturer case studies	Directional only, never quote as a projection

The Del Mar High School figures in Section 5 are Tier 3, published by a vendor. They are included because they are the only school-specific labor numbers available, and they are flagged as such wherever used.

3. What the Technology Actually Is in 2026

Three separate product classes get called autonomous mowers. They are not interchangeable and they have very different cost structures.

Class 1: Boundary-free robotic mowers

These are the descendants of the residential robot mower. They live outdoors permanently in a charging dock, run on their own schedule day and night, and cut a small amount of grass very frequently rather than a lot occasionally. Modern units navigate by RTK satellite positioning rather than a buried perimeter wire, which is what made them practical for institutional use. Setup is a matter of hours, not days.

- Kress KR233 RTK, about $8,500, recommended for 3 acres and rated to 4.5 acres.
- Kress KR236 RTK, about $13,800, recommended for 6 acres and rated to 9 acres.
- Husqvarna CEORA 546 EPOS starter kit, $32,829.94, rated up to 18 acres at twice-weekly mowing quality or up to 6 acres at continuous sports-turf quality.
- ECHO TM-2050, rated to about 15 acres, quoted rather than list priced.

The important behavioral difference is that these machines mulch tiny clippings continuously. There is no bagging, no windrow, and no cleanup. The turf generally looks better than it does under a weekly

cut. The tradeoff is that they are slow, they are always out there, and they cannot handle a lawn that has been left to grow tall.

Class 2: Autonomy retrofits on conventional commercial mowers

Here a normal commercial zero-turn or stand-on mower is fitted with sensors and software so it can drive itself while a human supervises from nearby. Greenzie is the best known example, sold on Wright Manufacturing machines at $33,999 for the mower plus $12,000 per year for the robotic software activation. This class cuts at conventional speed and handles tall grass, but the software subscription is a permanent operating cost, and the machine still requires a person on site.

Class 3: Purpose-built autonomous commercial mowers

Scythe Robotics is the leading example. The M.52 is a 52 inch side-discharge machine, roughly 1,425 pounds, that mows 1.5 to 2 acres per hour at about 6.5 mph, runs a full workday on a charge, and recharges in 6 to 8 hours on 240 volt. It carries eight cameras and twelve ultrasonic sensors. It is sold only as robots as a service, with a quoted base lease plus a per-autonomous-acre charge, reported at around $30 per acre. Machine, service, parts, repairs, and software updates are included. Travel time and charging time are not billed.

Two constraints matter for a school district. Scythe operates on supervised autonomy, meaning a crew member has to be nearby, and it is not designed for intricate or obstacle-dense properties. It also had regional availability limited to roughly 30 states as of 2025, so Tennessee coverage needs to be confirmed with the vendor before this class is modeled seriously.

4. Vendor Landscape and Published Pricing

Vendor and model	Class	Published price	Rated capacity	Commercial model
Kress KR233 RTK	1	$8,499.99	3 ac rec, 4.5 ac max	Purchase
Kress KR236 RTK	1	$13,799.99	6 ac rec, 9 ac max	Purchase
Kress KR800 Voyager	1 / 3	$60,000+	High output	Purchase
Husqvarna CEORA 546 EPOS	1	$32,829.94 starter kit	18 ac standard, 6 ac sports	Purchase, Lease Plus available
ECHO TM-2050	1	Quote only	Up to 15 ac	Purchase
Greenzie on Wright	2	$33,999 + $12,000/yr	Conventional mower rate	Purchase plus annual software
Scythe M.52	3	Quote, about $30/ac	1.5 to 2 ac/hr	RaaS only
RC Mowers AMR	3	Quote only	Large open areas	Purchase, Sourcewell contract

Prices are as published in 2025 and 2026 dealer and trade-association sources. Treat every one as a starting point for a quote, not a budget line. Confirm current pricing before any figure reaches a board packet.

Recurring costs that are easy to miss

- Site setup and mapping, $500 to $3,000 or more depending on property complexity.

- Blade replacement, which is frequent on Class 1 machines because the small blades are consumable by design.
- Battery replacement, generally a multi-year reserve item rather than an annual one.
- Software subscription, which is $12,000 per year on the Greenzie model, and reportedly zero on Husqvarna and ECHO.
- RTK reference station, included in the CEORA starter kit but a separate line on some competitors.
- Electricity for charging, which is real but small and displaces gasoline.

5. What Schools Have Actually Done

Del Mar High School, California, 2025

The single most useful school data point available, and the only one with before and after labor hours. One NEXMOW M2 was deployed on the approximately 1.2 acre main front lawn plus athletic field areas. Before deployment, mowing consumed two to three staff members during peak growing season, reported as 26 to 30 staff hours per week. After deployment, the same ground required about 6 hours per week for oversight and battery management.

Other reported outcomes: no jobs were eliminated and the grounds crew shifted to other work, gasoline purchasing for those areas ended, deployment took a single afternoon with no buried wire because the unit uses GPS and RTK, and coaching staff could pause or reschedule zones through an app when practice fields were in use. No problems were documented, which is itself a signal that this is a vendor-published account rather than an independent evaluation.

Tier 3 source. The labor hour reduction is plausible and directionally consistent with the technology, but a roughly 80 percent labor reduction claim published by the manufacturer should be validated in a GCS pilot before it is used in a projection.

Higher education and municipal adoption

College campuses and municipal parks departments are the current volume buyers, driven by groundskeeping labor shortages rather than by cost savings alone. One facility manager account reports mowing costing about $1,000 per acre annually and nine autonomous mowers saving roughly $10,000 per year, which implies the fleet displaced about ten acres of conventional mowing. That is a Tier 3 figure and the underlying cost accounting is not published.

Athletic fields specifically

Sports turf is the strongest use case in the whole category and the most relevant one for GCS given Greeneville High and Greeneville Middle athletics. Continuous light mulching mowing produces more uniform height than a weekly cut, and uniform height is a genuine playing-surface and injury-risk issue, not a cosmetic one. Striping direction and speed can be set in the app on several platforms, and at least one unit, the FJD RM-21, both cuts and paints field lines.

The known operational catch is that repeatedly running mower wheels over painted lines accelerates paint wear and creates visible tracking. On a robot that runs continuously this compounds faster than it does with a weekly cut. Any athletic field deployment needs a route plan that varies wheel paths across

painted areas, and the athletic director and field crew need to be in the pilot design conversation from the start.

6. The Labor Question

This is the part that determines whether the project is a savings project or an efficiency project, and it should be settled before any equipment is priced. The two paths produce completely different financial statements and completely different conversations with staff.

	Reduce headcount	**Redeploy hours**
Where savings appear	Payroll line, immediately	Deferred maintenance, overtime, contract work
Size of savings	Large and easy to show	Smaller and harder to prove
Staff reaction	Adversarial	Generally supportive
Realistic in current market	Only through attrition	Yes
Board narrative	Cutting jobs to buy robots	Getting more out of the crew we can hire

Every school district found in this review chose redeployment, and the districts that documented their reasoning framed automation as a response to positions they could not fill rather than positions they wanted to eliminate. That framing is not just politics. In a labor market where grounds and custodial vacancies go unfilled for months, a machine that covers an unfilled position is worth more than a machine that displaces a filled one, and it carries none of the labor-relations cost.

For GCS the practical version is this. If the district cannot currently keep grounds coverage at the level it wants during peak growing season, autonomous mowing buys back capacity. If grounds coverage is fully staffed and adequate, the savings have to come from an actual reduction in hours or contract spend, and that reduction has to be identified specifically before purchase or it will not materialize.

7. What Drives the Payback Math

The accompanying workbook, Autonomous Mower ROI Model.xlsx, implements this model with live formulas. The variables below are the ones that actually move the answer, in rough order of leverage.

1. Mowable acres per site. Nothing matters more. A robot rated to 6 acres that is deployed on 1.5 acres has an effective cost per acre four times higher than its spec sheet implies.
2. Current labor hours spent mowing that specific ground, per week, at peak. This is the number that converts to dollars. District-wide grounds hours are not a substitute because trimming, edging, beds, and athletics prep do not go away.
3. Loaded hourly labor rate, meaning wage plus benefits plus payroll tax plus workers compensation. Published industry examples use around $22.50 per hour loaded, but GCS should use its own classified rate.
4. Length of the mowing season. In upper East Tennessee this is roughly 30 to 34 weeks, which is materially better than northern benchmarks and improves payback.

5. Cuts per week under the current regime. A site mowed twice a week has twice the displaceable labor of one mowed weekly.
6. Obstacle density. Every tree ring, sign post, bench, and drainage feature either slows the machine or requires a person with a trimmer, which caps the achievable labor reduction well below 100 percent.
7. Fuel and small-engine maintenance currently consumed on that ground. Real but secondary. Published examples put fuel at roughly $3.68 per mowing hour at $3.50 per gallon.

A worked illustration, not a GCS projection

To show the shape of the math with a mid-size Class 1 unit. Assume a 6 acre open athletic and lawn area currently mowed twice weekly for 8 hours per week total, over a 32 week season, at $22.50 per hour loaded. That is 256 hours per year, or $5,760 of labor. Add roughly $1,000 of fuel and small-engine maintenance and the displaceable annual cost is about $6,760. Against a $13,800 machine plus $1,500 setup, simple payback lands near 2.3 years if the labor is genuinely eliminated, and closer to never if the hours simply move to other work without reducing overtime or contract spend.

The $22.50 loaded rate and the fuel figure are industry benchmarks, not GCS figures. Replace both with district actuals in the workbook. The 8 hours per week is an assumption for illustration only. No GCS mowing hours are on file.

8. What Breaks It

The failure modes below are the ones most likely to turn a good business case into a machine sitting in a maintenance bay.

Terrain

Standard two-wheel-drive robot mowers are generally suitable only for slopes under about 35 percent grade. Above that an all-wheel-drive model is required, and some AWD and tracked units are rated to 70, 80, and even 100 percent grade. Upper East Tennessee campuses commonly have banks, swales, and detention basins well past the 2WD limit. Slope has to be measured, not eyeballed, before a model is selected. Wet grass compounds the problem because it reduces traction, and this region gets substantial summer rainfall.

Tree cover and RTK signal

RTK satellite positioning is the technology that made these machines practical, and it is easily blocked by tree canopy. A shaded quad or a lawn ringed with mature trees can degrade or lose positioning entirely. Models with LiDAR or visual SLAM as a fallback keep working under canopy, and that capability should be a hard requirement for any GCS site with meaningful tree cover rather than a nice-to-have.

Weather and season

Most units are weather-resistant but degrade in heavy rain and extreme temperatures. There is also a storage and winterization requirement that adds a small amount of labor back at both ends of the season.

Security and mischief

Theft protection is standard across the category and appears adequate. Units carry GPS tracking, geofencing with alerts, PIN codes, and alarms, and there is at least one documented case of GPS data leading police directly to a stolen unit. The more realistic risk on a school campus is not theft but interference. Vendors report that the most common damage is accidental, caused by debris left on the turf, and a school campus generates more of that than a cemetery or an office park does. Students testing what the robot does when they block it is a predictable behavior and an argument for running outside school hours during the academic year.

Student safety and district liability

No published guidance specific to autonomous mowers operating on occupied K-12 campuses was found in this review. That is a gap, not a clean bill of health. Before deployment the district should confirm coverage treatment with the insurance carrier, since this is an unmanned powered machine operating on property where children are present, and GCS carries property and liability through arrangements that should be checked rather than assumed. The conservative operating rule, and the one Del Mar effectively used, is that the machine runs when students are not on that ground, with app-based zone pausing when athletics need the field.

The obstacle-density trap

The single most common way this investment disappoints is buying capacity the property cannot use. A machine rated for 18 acres of clean open turf will not deliver 18 acres of a school campus, because school campuses are full of sidewalks, bus loops, playground equipment, portable classrooms, sign monuments, and landscape beds. Rate the ground, not the machine.

9. Procurement Path for GCS

Threshold

Greene County had a 2020 census population of 70,152, which is above the 40,000 population line in state purchasing law. Under TCA 49-2-203, an LEA in a county of 40,000 or more may make purchases of less than $25,000 in the open market without newspaper advertisement, but must base them on at least three competitive bids whenever possible. Purchases at or above the applicable threshold require competitive sealed bids solicited by newspaper advertisement, unless the district follows the local governing body's purchasing procedures instead.

The practical consequence is favorable for a pilot. A single Class 1 machine in the $8,500 to $14,000 range, plus setup, sits comfortably under $25,000 and can be handled with three quotes. A district-wide fleet or a CEORA-class deployment crosses the threshold and needs a bid or a cooperative contract.

Confirm the current threshold figure with the district CFO before it appears in any board document. The threshold is set by reference to TCA 12-3-1212 and has been amended by the legislature more than once, most recently in the 2021-2022 session raising it from $10,000 to $25,000 for LEAs in counties of 40,000 or more.

Cooperative contracts

Cooperative purchasing avoids the bid process entirely by riding a contract another agency already competed. RC Mowers holds Sourcewell contract 112624-EMB in the Grounds Maintenance Equipment and Related Attachments category, which covers robotic and autonomous mowers and is available to public agencies including school districts. Sourcewell, OMNIA Partners, and NASPO ValuePoint are the three main programs to check. If GCS is not already a Sourcewell participating agency, registration is free and worth doing regardless of the outcome here.

Purchase versus lease versus service

Model	Fits when	Watch for
Purchase	Capital funds available, long expected life, simple machine	Battery and obsolescence risk sit with the district
Lease (Husqvarna Lease Plus)	Preserving capital, wanting a fixed operating line	Total paid over term exceeds purchase price
RaaS (Scythe, per acre)	Uncertain acreage, wanting service included, avoiding capital request	Cost scales forever with use, never converts to an asset
Purchase plus software subscription (Greenzie)	Wanting conventional mower speed with autonomy	$12,000 per year is permanent and independent of use

For a first GCS deployment, purchase of a single Class 1 unit is the simplest path. It stays under the bid threshold, it creates an asset, it produces a full season of district-specific data, and it does not commit the district to a recurring line item before the value is proven.

10. Recommended Pilot Design

The goal of a pilot is not to save money in year one. It is to produce GCS numbers that replace the benchmark assumptions in the workbook, so that a second-year decision can be made on evidence rather than on vendor material.

8. Pick one site with the most contiguous, open, low-slope turf GCS owns. Measure it. Do not estimate it. Aerial measurement from the county GIS parcel viewer or a simple wheel-and-map exercise is sufficient.
9. Baseline before buying anything. For four consecutive weeks in peak growing season, log actual mowing hours on that specific ground, separately from trimming, edging, and bed work. This is the single most valuable thing the district can do, and it costs nothing but a clipboard.
10. Measure slope at the steepest section and confirm the selected model's rating exceeds it with margin. Confirm tree canopy coverage and require LiDAR or visual fallback if canopy is significant.
11. Get three quotes on a single Class 1 unit sized to the measured acreage, plus setup. Check the Sourcewell RC Mowers contract as a fourth reference point even if the district does not buy from it.
12. Confirm insurance treatment with the carrier in writing before the machine operates on occupied ground.
13. Set the operating window so the machine does not run on turf while students are on it during the school day, with app zone pausing available to coaches and the athletic director.

14. Run one full season. Log oversight hours, downtime, incidents, blade and consumable spend, and any interference events.
15. Compare logged season-two labor against the pre-purchase baseline. Only then extend or stop.

11. Data GCS Must Collect Before Any Projection

None of the following exists in district files today. The The district operations tracker records square footage maintained and square footage cleaned as open items requested from operations in May 2026 and still outstanding. Until these are supplied, any dollar projection is an assumption dressed as an analysis.

Data point	Owner	Why it matters
Mowable acres by site, measured	Operations director	Primary driver of the entire model
Slope at steepest mowed section by site	Operations director	Determines 2WD versus AWD, or disqualifies the site
Tree canopy over mowed areas	Operations director	Determines whether RTK-only models are viable
Weekly mowing hours at peak, by site	Operations director	Converts acreage into displaceable dollars
Cuts per week by site	Operations director	Doubles or halves the labor base
Loaded hourly rate, grounds classification	HR director / CFO	Wage plus benefits plus tax plus comp
Current contract mowing spend, if any	CFO	Contract dollars are easier to capture than internal hours
Annual fuel and small-engine maintenance for mowing	Operations director	Secondary savings line
Current grounds vacancies and time to fill	HR director	Determines whether this is a savings or a capacity case

The Classified Salary Study for 2026-2027 is active and will affect the loaded rate input. Coordinate the labor rate figure with that work rather than pulling a rate independently.

12. Verdict

Autonomous mowing is a real and reasonably mature technology with a defensible use case for a district like Greeneville, concentrated on athletic fields and large open lawns rather than on general campus grounds. The equipment cost for a meaningful pilot is modest, under the state open-market threshold, and reversible.

The honest constraint is that GCS is not currently in a position to say what it would save, because the district does not know its own mowable acreage or its own mowing hours. That is a two-week fix, not a two-year one, and it should happen before any vendor conversation. A district that walks into a quote knowing its acreage, its slopes, its canopy, and its current hours will get a better machine at a better price than one that walks in asking what is available.

Recommended next step: request the acreage and hours data from the operations director, run the four-week mowing time study at the largest open site, and populate the accompanying workbook. Revisit the purchase question with real numbers in hand.

13. Sources

Tier 1: Government, board record, and independent reporting

1. US Census Bureau QuickFacts, Greene County, Tennessee (2020 population 70,152) https://www.census.gov/quickfacts/greenecountytennessee

2. Tennessee Code 49-2-203, Duties and powers, LEA purchasing thresholds https://law.justia.com/codes/tennessee/title-49/chapter-2/part-2/section-49-2-203/

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5. RC Mowers awarded Sourcewell cooperative contract (press release) https://www.prnewswire.com/news-releases/rc-mowers-awarded-sourcewell-cooperative-contract-302399190.html

Tier 2: Manufacturer specifications and published dealer pricing

1. Husqvarna CEORA commercial robotic mower, specifications and kit contents https://www.husqvarna.com/us/discover/ceora/

2. Green Industries Equipment, commercial robotic mower cost and ROI guide (Kress pricing, setup cost ranges) https://www.greenindustriesequipment.com/blog/how-much-does-a-commercial-robotic-mower-cost/

3. National Association of Landscape Professionals, pricing structures and maintenance of commercial autonomous mowers (CEORA kit price, Greenzie pricing, maintenance intervals) https://blog.landscapeprofessionals.org/pricing-structures-and-maintenance-of-commercial-autonomous-mowers/

4. MowScout, Scythe M.52 specifications and RaaS structure https://mowscout.com/guides/scythe-m52-robot-mower

5. Rural Lifestyle Dealer, examining Scythe Robotics pay-as-you-go rental model https://www.rurallifestyledealer.com/articles/10820-examining-scythe-robotics-pay-as-you-go-rental-model-for-its-electric-autonomous-mowers

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7. RC Mowers, autonomous mowing robots product line https://www.rcmowersusa.com/robotic-mowers/autonomous-mowing-robot/

Tier 3: Vendor and vendor-adjacent material, directional only

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2. PUPN Magazine, the rise of robotic mowers in campus grounds management https://pupnmag.com/article/the-rise-of-robotic-mowers-revolutionizing-campus-grounds-management/

3. Mammotion, commercial robotic lawn mower price guide (savings and payback ranges) https://us.mammotion.com/blogs/news/commercial-robotic-lawn-mowers-price-guide

4. Mowing Magic, commercial robotic mower prices https://mowingmagic.com/commercial-robotic-mower-prices/

5. FieldCamp, how to price commercial lawn care jobs 2026 (per-acre contract rates) https://fieldcamp.ai/lawn-care-software/how-to-price-commercial-lawn-care-jobs/

6. Eden, mowing cost breakdown (loaded labor and fuel per hour) https://eden.studio/blog/how-much-to-charge-for-mowing-2-acres/

7. GST Robotic Mowers, maximum slope for robotic mowers https://gstroboticmowersinc.com/what-is-the-maximum-slope-for-a-robot-lawn-mower/

8. Mammotion, robotic mowers without perimeter wire, RTK and canopy limitations https://eu.mammotion.com/blogs/news/robotic-lawn-mowers-without-perimeter-wire-tested

9. Belrobotics, theft and vandalism of robotic mowers https://www.belrobotics.com/en/blog/faq/what-about-theft-or-vandalism-of-robotic-mowers/

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12. RC Mowers, sports field mowing applications https://www.rcmowersusa.com/robotic-mowers/autonomous-mowing-robot/applications/sports-field-mowing/

Internal GCS sources

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2. GCS Operations Infographic Data.xlsx, maintenance work orders 1,545 for 7/1/25 to 5/26/26